AUDIT REPORT

predictive-maintenance: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/predictive-maintenance
Numbers taken from commit	`70531ff3190a5ff2de81b5edc3ef1ee5888ae47c` (outputs last changed in `70531ff (2026-10-13 21:58:39 +0400)`)
Video	`docs/demo/demo.mp4` · 86.0 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:37
Automated checks	62 pass · 0 fail · 1 manual (of 63 figures)
Assumptions	10 (A-01 to A-10)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/*.csv + outputs/results.json, closed_loop_runs.csv, work_orders.json (~2 min)
.venv/bin/python docs/demo/source/tools/export_work_orders.py docs/demo/source/assets/plant_run.json   # ~25 s
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/plant_run.json docs/demo/source/compositions --rows 0,1,2,3,4,5
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/predictive-maintenance-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/predictive-maintenance-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem' (problems 1-3)	README lines 10-22	-	general knowledge	✓ verified
F-03	200 machines	02, 05, 08	data/assets.csv	row count	-	direct	✓ verified
F-04	About 6 planned jobs a week / crew 6 jobs / week	02, 08	outputs/results.json	crew_jobs_per_week	-	direct	✓ verified
F-05	110 pumps · 40 compressors · 50 heat exchangers	02	data/assets.csv	asset_type counts	-	direct	✓ verified
F-06	219,000 daily sensor readings / .csv.gz · 219,000	02, 05	data/ sensor_readings.csv.gz	row count	-	direct	✓ verified
F-07	130–157 breakdowns a year on the calendar	02	data/ maintenance_log.csv	kind=corrective per year: 143, 130, 157	min..max of yearly corrective counts	derived	✓ verified
F-08	SENSORS · SERVICE LOG · RISK MODEL · WORTH A JOB? · CREW PLAN · WORK ORDERS	03	src/ predictive_maintenance /	functions	-	direct	✓ verified
F-09	fewer breakdowns / lower cost / higher uptime	03	outputs/results.json closed_loop	gbm-sensors vs fixed-interval at crew 6	failures, cost_total lower; uptime higher	derived	✓ verified
F-10	EVERY WORKING DAY	03	src/ predictive_maintenance /config.py:107	working_days = (0..4)	-	direct	✓ verified
F-11	Replay the plant to today, then fork it	04	src/ predictive_maintenance /plant.py:281-293; tests/test_plant.py:48	fork(); test_fork_gives_every_policy_the_same_future	-	direct	✓ verified
F-12	Each policy runs a full year	04	src/ predictive_maintenance /cli.py (run_policy(..., DAYS_PER_YEAR, ...)); config.py:13	DAYS_PER_YEAR = 365	-	direct	✓ verified
F-13	tick barcodes (no numbers)	04	demo assets/plant_run.json (export_work_orders.py); outputs/closed_loop_runs.csv	LOOP block in compositions/explainer.html	number of ticks per branch = failures for seed 101, crew 6	derived	✓ verified
F-14	0.779	04, 09	outputs/results.json	offline.policies.cox-sensors.c_index	-	direct	✓ verified
F-15	0.753	04	outputs/results.json	offline.policies.gbm-sensors.c_index	-	direct	✓ verified
F-16	29%	04	outputs/results.json closed_loop	cox-sensors, crew 6, failures_caught_share (0.2929)	x100, rounded	direct	✓ verified
F-17	69%	04	outputs/results.json closed_loop	gbm-sensors, crew 6, failures_caught_share (0.6868)	x100, rounded	direct	✓ verified
F-18	seed 7 · 3 years	05	src/ predictive_maintenance /cli.py (--data-seed default 7); config.py:14	HISTORY_YEARS = 3	-	direct	✓ verified
F-19	.csv · 1,005 jobs	05	data/ maintenance_log.csv	row count	-	direct	✓ verified
F-20	eval only · 1,205	05	data/lives_truth.csv	row count	-	direct	✓ verified
F-21	observable only	05	src/ predictive_maintenance /features.py (_labels); tests/test_features.py :33	test_labels_drop_rows_a_preventive_job_hides	-	direct	✓ verified
F-22	plant.py, features.py, models.py (LightGBM, Cox, Weibull, AFT), config.py, policy.py (CostAwareModel, schedule, other policies), workorders.py, simulate.py, evaluate.py	05	src/ predictive_maintenance /	classes and functions	-	direct	✓ verified
F-23	60-day window	05	src/ predictive_maintenance /config.py:19	WINDOW_DAYS = 60	-	direct	✓ verified
F-24	26 per asset-day	05	src/ predictive_maintenance /features.py:20-28	5 sensors x 4 + load_m7 + load_rel3 +	5 x 4 + 3 + 3 = 26	derived	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
			(FEATURE_NAMES)	days_since_maint + 3 type flags			
F-25	p(fail ≤ 14 d) / fail within 14 d / p(14 d)	05, 06	outputs/results.json	horizon_days	-	direct	✓ verified
F-26	119 trees	05	outputs/results.json	training.best_iteration	-	direct	✓ verified
F-27	T* 187·121·218 d	05	outputs/results.json	fixed_intervals_days	-	direct	✓ verified
F-28	saving = p·C_fail − C_job; > 0 ? yes: propose	05	src/predictive_maintenance/policy.py:109-110	saving = p * FAILURE_COST - PLANNED_COST; propose where saving > 0	-	direct	✓ verified
F-29	≤ 2 a day · ≤ 6 a week · 21-day cooldown	05	src/predictive_maintenance/policy.py:144-145; config.py:20, :106-107	per_day = ceil(6 / 5); COOLDOWN_DAYS = 21	ceil(6/5) = 2	derived	✓ verified
F-30	7 policies × 5 forks	05, 10	outputs/closed_loop_runs.csv	distinct policies at crew 6; distinct seeds	-	direct	✓ verified
F-31	make run ≈ 2 min / ~2 min	05, 10	outputs/results.json	runtime_seconds = 127.3	127.3 / 60 = 2.1 min	derived	✓ verified
F-32	Planner · accept → CMMS	05	src/predictive_maintenance/workorders.py:1-5 (docstring); README 'Known limitations' (schema mapping not done)	-	-	direct	✓ verified
F-33	daily vibration · last 60 days (line chart)	06	data/sensor_readings.csv.gz ; demo assets/plant_run.json trace	asset_id = COMP-141, 2024-11-01..2024-12-30	values compared to 3 decimals	direct	✓ verified
F-34	alert · 2024-12-31	06	outputs/work_orders.json	[0].created_on	-	direct	✓ verified
F-35	+52% vs its baseline / vibration +52%	06, 07	outputs/work_orders.json	[0].evidence[0] = vibration, 0.521	x100, rounded	direct	✓ verified
F-36	CBM-2024-00001 · COMP-141 · U200-K-141 · 0.736 · P1 · 2025-01-03 · rotating-equipment	06	outputs/work_orders.json	[0]	-	direct	✓ verified
F-37	hidden life ended 1.8 days later (bearing)	06	demo assets/plant_run.json work_orders[0].truth (Plant.latent_remaining at the alert, fork seed 101); corroborated by data/lives_truth.csv (the same life, in the history run, ended by a bearing failure on day 731)	latent_days_to_failure, latent_mode	-	direct	✓ verified
F-38	232 · 138 · 111 · 56 · 43 · 3	06	outputs/closed_loop_runs.csv (seed 101, crew 6); curves from demo assets/plant_run.json	failures	-	direct	✓ verified
F-39	fork 1 of 5	06	src/predictive_maintenance/config.py:108; demo assets/plant_run.json seed	eval_seeds[0] = 101	-	direct	✓ verified
F-40	CBM-2024-00001 … CBM-2025-00006	07	outputs/work_orders.json [0:6], injected into compositions/value.html	work_order_id, asset_id, failure_probability, evidence[0], expected_saving, due_by, crew	p to 2 dp; saving / 1000 to 1 dp; crew shortened (rotating-equipment → rotating)	direct	✓ verified
F-41	+ 6 more orders the following week	07	outputs/results.json	work_orders_first_two_weeks = 12	12 - 6	derived	✓ verified
F-42	All 6 pass validate()	07	src/predictive_maintenance/cli.py (assert not bad); tools/export_work_orders.py asserts the same	validate()	-	direct	✓ verified
F-43	5 of 6 were under 14 days from failing	07	demo assets/plant_run.json work_orders[0:6].truth	latent_days_to_failure: 1.8, 0.3, 5.0, 11.0, 3.3, 18.1	count(0 < d <= 14)	derived	✓ verified
F-44	CREW 6 / 6	07	outputs/work_orders.json	orders created 2024-12-30..2025-01-03	count	derived	✓ verified
F-45	TUE 31 DEC · WED 1 JAN · THU 2 JAN	07	outputs/work_orders.json	created_on of orders 1-6	-	direct	✓ verified
F-46	99.47%	08	outputs/results.json closed_loop	gbm-sensors, crew 6, uptime	x100	direct	✓ verified
F-47	vs calendar 99.00%	08	outputs/results.json closed_loop	fixed-interval, crew 6, uptime	x100	direct	✓ verified
F-48	+0.47 pp	09	outputs/results.json	paired_gbm_minus.fixed-interval.uptime_pp_mean	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-49	44.2	08	outputs/results.json closed_loop	gbm-sensors, crew 6, failures	-	direct	✓ verified
F-50	calendar 111.2	08	outputs/results.json closed_loop	fixed-interval, crew 6, failures	-	direct	✓ verified
F-51	2.02M	08	outputs/results.json closed_loop	gbm-sensors, crew 6, cost_total = 2,015,700	/ 1e6	direct	✓ verified
F-52	4.01M	08	outputs/results.json closed_loop	fixed-interval, crew 6, cost_total = 4,009,400	/ 1e6	direct	✓ verified
F-53	68.7%	08	outputs/results.json closed_loop	gbm-sensors, crew 6, failures_caught_share	x100	direct	✓ verified
F-54	median lead 7.7 days	08	outputs/results.json closed_loop	gbm-sensors, crew 6, median_lead_days	-	direct	✓ verified
F-55	the model beat every runnable policy in all 5 forks	08	outputs/results.json	paired_gbm_minus.*.seeds_gbm_better_uptime	all == 5	direct	✓ verified
F-56	+0.30 pp	09	outputs/results.json closed_loop	alarm-limits uptime - fixed-interval uptime (crew 6)	(0.992978 - 0.989989) x 100 = 0.299	derived	✓ verified
F-57	+0.19 pp	09	outputs/results.json closed_loop	cox-sensors uptime - fixed-interval uptime (crew 6)	(0.991939 - 0.989989) x 100 = 0.195	derived	✓ verified
F-58	+0.66 pp	09	outputs/results.json closed_loop	oracle uptime - fixed-interval uptime (crew 6)	(0.996549 - 0.989989) x 100 = 0.656	derived	✓ verified
F-59	38 of the 44 breakdowns left are sudden	09	outputs/ closed_loop_runs.csv	gbm-sensors, crew 6: mean sudden_failures 37.6, mean failures 44.2	rounded to whole numbers	derived	✓ verified
F-60	The model caught 1 of 189	09	outputs/results.json	gbm_caught_by_mode.sudden	-	direct	✓ verified
F-61	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-62	LightGBM · lifelines · closed-loop sim · Python	10	pyproject.toml dependencies; src/predictive_maintenance/simulate.py	lightgbm, lifelines	-	direct	✓ verified
F-63	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 14:37)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	200	200	✓ pass
F-04	6	6	✓ pass
F-05	110 pumps · 40 compressors · 50 heat exchangers	110 pumps · 40 compressors · 50 heat exchangers	✓ pass
F-06	219,000	219,000	✓ pass
F-07	130–157	130–157	✓ pass
F-08	true	true	✓ pass
F-09	true	true	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	true	true	✓ pass
F-14	0.779	0.779	✓ pass
F-15	0.753	0.753	✓ pass
F-16	29%	29%	✓ pass
F-17	69%	69%	✓ pass
F-18	true	true	✓ pass
F-19	1,005	1,005	✓ pass
F-20	1,205	1,205	✓ pass
F-21	true	true	✓ pass
F-22	true	true	✓ pass
F-23	true	true	✓ pass
F-24	26	26	✓ pass
F-25	14	14	✓ pass
F-26	119	119	✓ pass
F-27	187·121·218	187·121·218	✓ pass
F-28	true	true	✓ pass
F-29	true	true	✓ pass
F-30	7 policies × 5 forks	7 policies × 5 forks	✓ pass
F-31	~2 min	~2 min	✓ pass
F-32	true	true	✓ pass
F-33	true	true	✓ pass
F-34	2024-12-31	2024-12-31	✓ pass
F-35	vibration +52%	vibration +52%	✓ pass
F-36	CBM-2024-00001 · COMP-141 · U200-K-141 · 0.736 · P1 · 2025-01-03 · rotating-equipment	CBM-2024-00001 · COMP-141 · U200-K-141 · 0.736 · P1 · 2025-01-03 · rotating-equipment	✓ pass
F-37	1.8 bearing	1.8 bearing	✓ pass
F-38	232 · 138 · 111 · 56 · 43 · 3	232 · 138 · 111 · 56 · 43 · 3	✓ pass

ID	Expected on slide	Recomputed from repo	Result
F-39	true	true	✓ pass
F-40	true	true	✓ pass
F-41	6	6	✓ pass
F-42	true	true	✓ pass
F-43	5 of 6	5 of 6	✓ pass
F-44	6 / 6	6 / 6	✓ pass
F-45	true	true	✓ pass
F-46	99.47%	99.47%	✓ pass
F-47	99.00%	99.00%	✓ pass
F-48	+0.47 pp	+0.47 pp	✓ pass
F-49	44.2	44.2	✓ pass
F-50	111.2	111.2	✓ pass
F-51	2.02M	2.02M	✓ pass
F-52	4.01M	4.01M	✓ pass
F-53	68.7%	68.7%	✓ pass
F-54	7.7	7.7	✓ pass
F-55	true	true	✓ pass
F-56	+0.30 pp	+0.30 pp	✓ pass
F-57	+0.19 pp	+0.19 pp	✓ pass
F-58	+0.66 pp	+0.66 pp	✓ pass
F-59	38 of 44	38 of 44	✓ pass
F-60	1 of 189	1 of 189	✓ pass
F-61	true	true	✓ pass
F-62	true	true	✓ pass
F-63	true	true	✓ pass

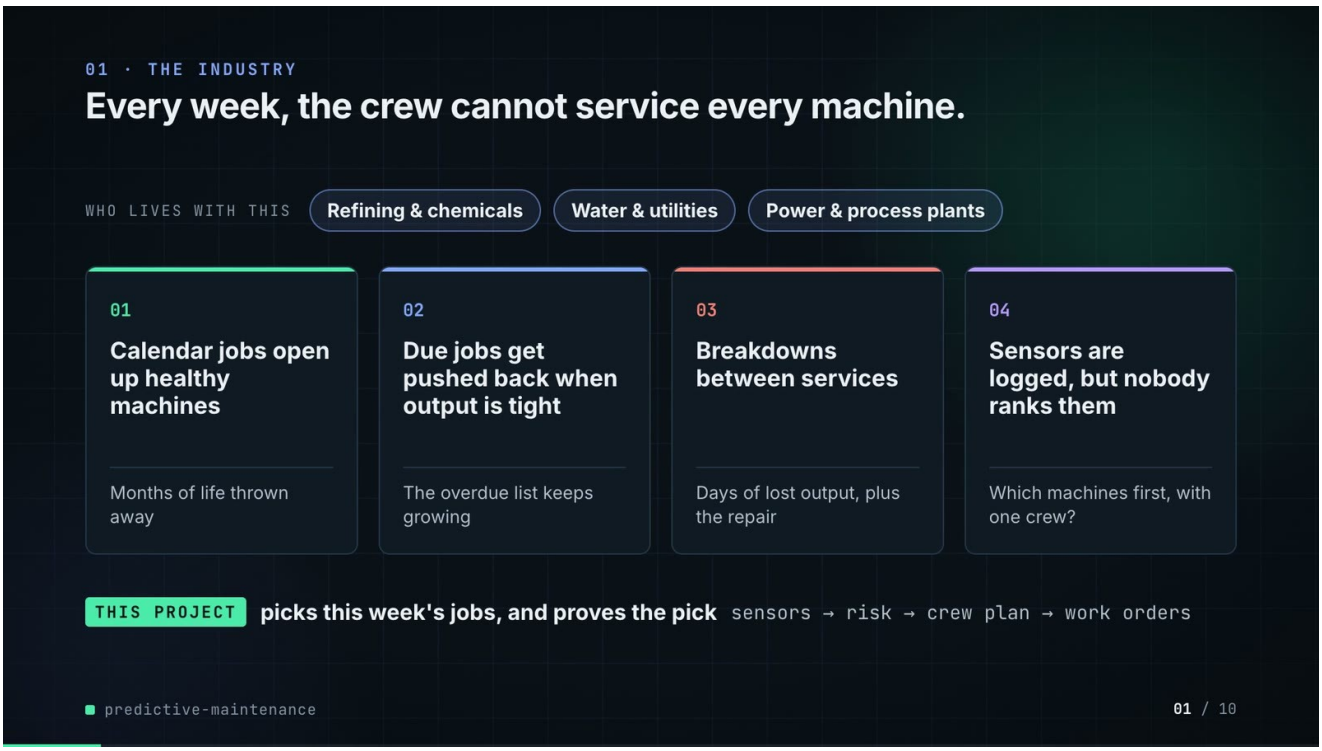
3. Assumptions register

ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic: a seeded day-by-day plant model (200 assets, 3 years) generates the sensor history, the maintenance log and the latent lives. The same plant model also runs the closed-loop evaluation, so results describe a plant with these physics, not a given real plant (README 'Known limitations', the biggest gap).	src/predictive_maintenance/plant.py; data.py:152-160 (generate, seed 7); config.py:14; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Failure precursors: each failure mode drifts the sensors for a mode-specific number of days before the failure (bearing 21 d, fouling 35 d, overheating 4 d, ...); sudden failures give no warning. These lengths are assumptions.	src/predictive_maintenance/config.py:93-101 (PRECURSOR_DAYS)	F-17, F-34, F-35, F-37, F-53, F-54, F-59, F-60	slides 04, 06, 07, 08, 09
A-03	Costs: a planned job costs 1,500 plus one day of downtime; a failure costs its repair (9k pump, 22k compressor, 12k heat exchanger) plus 3-5 days of downtime at 3,000 a day. One downtime cost for every asset.	src/predictive_maintenance/config.py:32-34, :61-62, :72-73, :83-84	F-28, F-40, F-51, F-52	slides 05, 07, 08
A-04	Crew budget: 6 planned jobs a week, Monday to Friday, at most ceil(6/5) = 2 a day; an asset serviced in the last 21 days is not offered again.	src/predictive_maintenance/config.py:106 (crew_jobs_per_week), :20 (COOLDOWN_DAYS); policy.py:144-145 (per_day)	F-04, F-29, F-44, F-45, F-46, F-49, F-51, F-53	slides 02, 05, 07, 08
A-05	Prediction horizon 14 days, feature window 60 days, baseline = window median excluding the last 14 days.	src/predictive_maintenance/config.py:18-19; features.py:18 (RECENT = 14)	F-23, F-25, F-34, F-35	slides 05, 06, 07
A-06	Every job is a perfect repair (the asset is as good as new); imperfect repair and infant mortality are not modelled.	README 'Known limitations'; plant.py (_new_life on service)	F-46, F-49, F-51, F-53	slides 08
A-07	Closed-loop evaluation: the plant is replayed to the end of the 2-year training history (2024-12-31), forked with seeds 101-105, and each policy runs 365 days from the same fork with the same random stream. Slides 08-09 show the mean of the 5 forks; slides 04 and 06 show fork 1 (seed 101).	src/predictive_maintenance/cli.py:146-168 (closed_loop); plant.py:281-293 (fork); config.py:108 (eval_seeds); docs/adr/0001-judge-policies-in-closed-loop.md	F-11, F-12, F-13, F-30, F-38, F-39, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-56, F-57, F-58, F-59	slides 04, 05, 06, 08, 09, 10
A-08	Truth for scoring: an alert is a catch when the asset's latent life would have ended within 14 days; the latent failure day comes from the plant model (lives_truth.csv / Plant.latent_remaining) and is never a feature.	src/predictive_maintenance/simulate.py:31-33 (true_alerts); features.py (latent_labels); data.py:1-9	F-16, F-17, F-37, F-43, F-53, F-60	slides 04, 06, 07, 08, 09
A-09	Video rounding: uptime to 2 decimals, percentages to whole numbers or 1 decimal as shown, costs in millions to 2 decimals, savings in thousands to 1 decimal, failure probability to 2 decimals on the sheet (3 on the work-order card), failure means to 1 decimal or whole numbers as shown.	this audit (display convention)	F-07, F-16, F-17, F-31, F-35, F-40, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-56, F-57, F-58, F-59	slides 02, 04, 05, 06, 07, 08, 09, 10
A-10	Slide 01 (industries, everyday problems) is general maintenance knowledge used as context; it is not a result of this repo and carries no numbers.	author's framing; README 'The problem' supports problems 1-3	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every week, the crew cannot service every machine.

WHO LIVES WITH THIS · Refining & chemicals · Water & utilities · Power & process plants

01 Calendar jobs open up healthy machines — Months of life thrown away

02 Due jobs get pushed back when output is tight — The overdue list keeps growing

03 Breakdowns between services — Days of lost output, plus the repair

04 Sensors are logged, but nobody ranks them — Which machines first, with one crew?

THIS PROJECT picks this week's jobs, and proves the pick · sensors → risk → crew plan → work orders

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem' (problems 1-3)` · README lines 10-22	-	✓

Assumptions: A-10

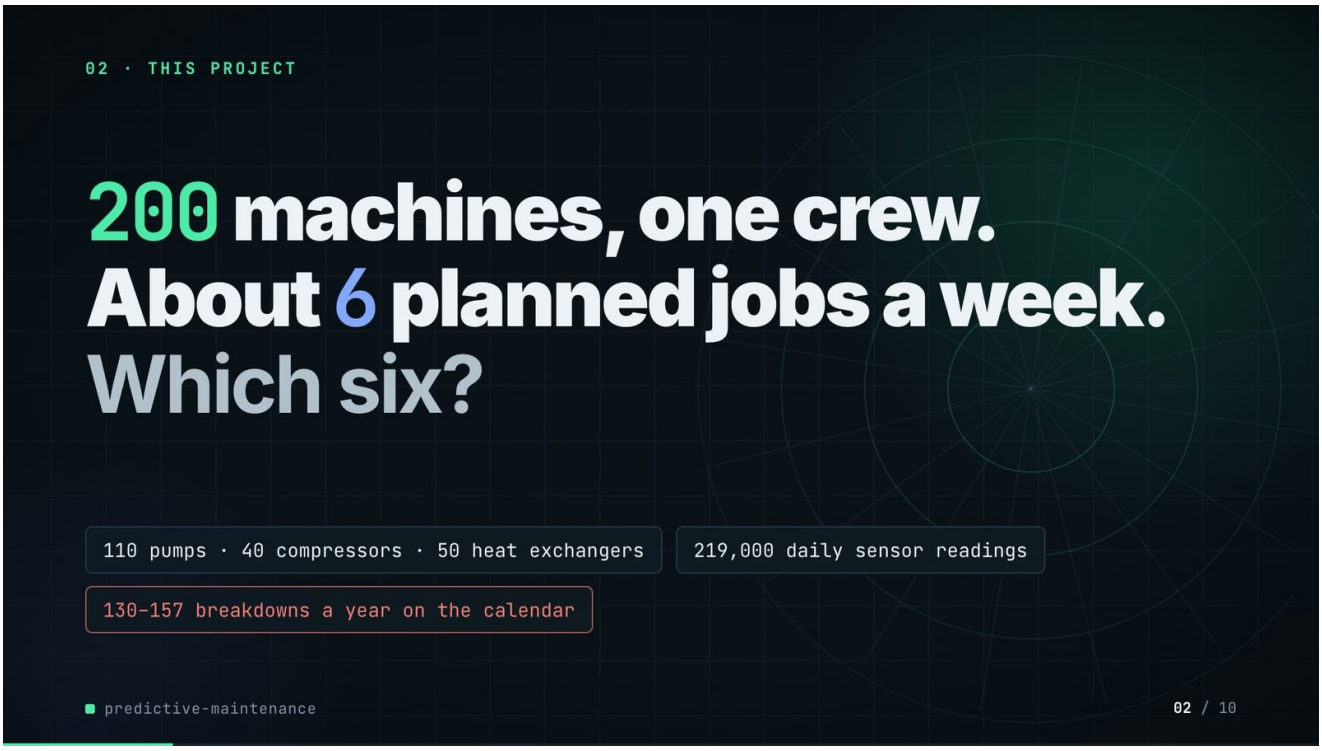
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

200 machines, one crew. About 6 planned jobs a week. Which six?

110 pumps · 40 compressors · 50 heat exchangers · 219,000 daily sensor readings · 130–157 breakdowns a year on the calendar

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	200 machines	`data/assets.csv` · row count	-	✓
F-04	About 6 planned jobs a week / crew 6 jobs / week	`outputs/results.json` · crew_jobs_per_week	-	✓
F-05	110 pumps · 40 compressors · 50 heat exchangers	`data/assets.csv` · asset_type counts	-	✓
F-06	219,000 daily sensor readings / .csv.gz · 219,000	`data/sensor_readings.csv.gz` · row count	-	✓
F-07	130–157 breakdowns a year on the calendar	`data/maintenance_log.csv` · kind=corrective per year: 143, 130, 157	min..max of yearly corrective counts	✓

Assumptions: A-01, A-04

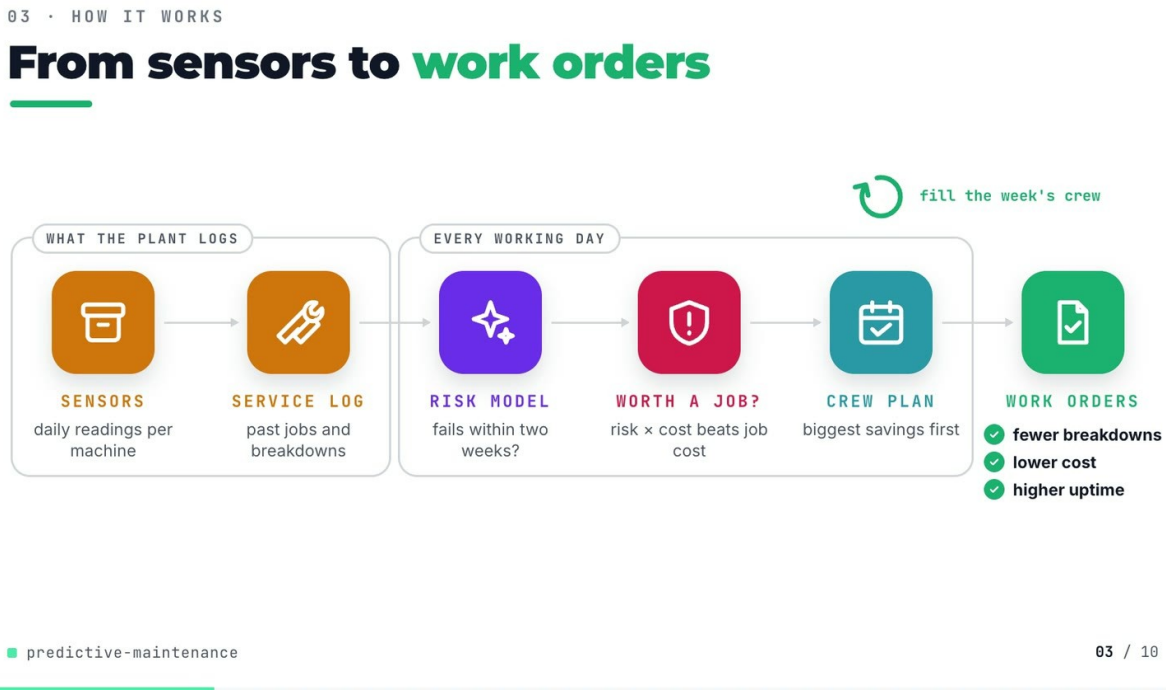
Illustrative, not data:

- Rings in the background are decoration.
- 'one crew': the scheduler treats the crew as one pool (README 'Known limitations').

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; plant size is config.py ASSET_TYPES.

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From sensors to work orders

[WHAT THE PLANT LOGS: SENSORS daily readings per machine → SERVICE LOG past jobs and breakdowns] → [EVERY WORKING DAY: RISK MODEL fails within two weeks? → WORTH A JOB? risk × cost beats job cost → CREW PLAN biggest savings first] → WORK ORDERS ✓ fewer breakdowns ✓ lower cost ✓ higher uptime

fill the week's crew (loop over the crew plan)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-08	SENSORS · SERVICE LOG · RISK MODEL · WORTH A JOB? · CREW PLAN · WORK ORDERS	`src/predictive_maintenance/` · functions	-	✓
F-09	fewer breakdowns / lower cost / higher uptime	`outputs/results.json closed_loop` · gbm-sensors vs fixed-interval at crew 6	failures, cost_total lower; uptime higher	✓
F-10	EVERY WORKING DAY	`src/predictive_maintenance/config.py:107` · working_days = (0..4)	-	✓

Assumptions: A-03, A-04, A-05

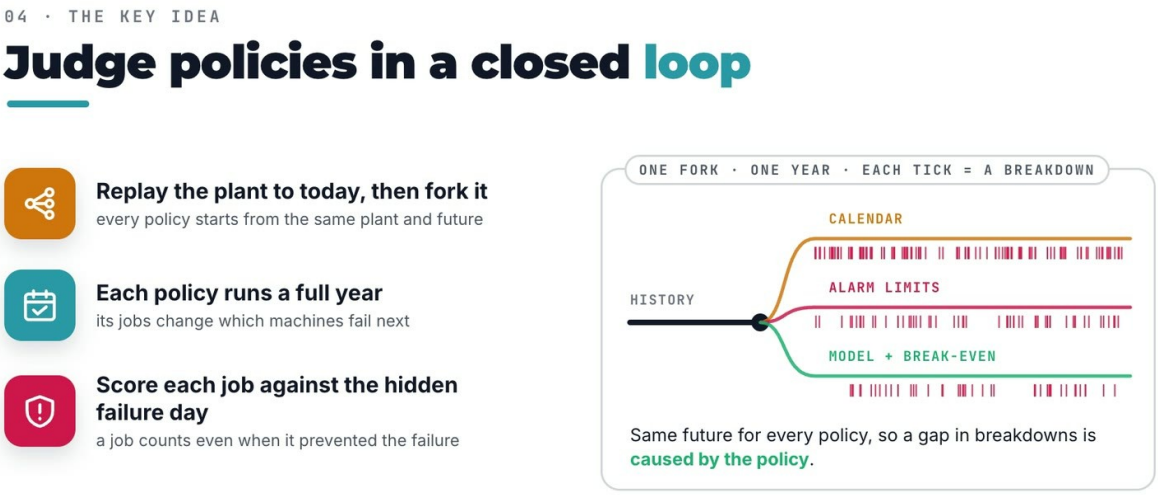
Illustrative, not data:

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth is F-09.

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition compositions/explainer.html (#ex-b ..., LOOP block injected by tools/inject_sheet.py)



On-screen text (verbatim)

04 · THE KEY IDEA

Judge policies in a closed loop

Replay the plant to today, then fork it — every policy starts from the same plant and future

Each policy runs a full year — its jobs change which machines fail next

Score each job against the hidden failure day — a job counts even when it prevented the failure

ONE FORK · ONE YEAR · EACH TICK = A BREAKDOWN: HISTORY → CALENDAR / ALARM LIMITS / MODEL + BREAK-EVEN

Same future for every policy, so a gap in breakdowns is caused by the policy.

Ranked by C-index alone, Cox would have won (0.779 vs 0.753). In the loop it caught 29% of failures, not 69%.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-11	Replay the plant to today, then fork it	src/predictive_maintenance/plant.py:281-293; tests/test_plant.py:48 · fork(); test_fork_gives_every_policy_the_same_future	-	✓
F-12	Each policy runs a full year	src/predictive_maintenance/cli.py (run_policy(..., DAYS_PER_YEAR, ...)); config.py:13 · DAYS_PER_YEAR = 365	-	✓
F-13	tick barcodes (no numbers)	demo assets/plant_run.json (export_work_orders.py); outputs/closed_loop_runs.csv · LOOP block in compositions/explainer.html	number of ticks per branch = failures for seed 101, crew 6	✓
F-14	0.779	outputs/results.json · offline.policies.cox-sensors.c_index	-	✓
F-15	0.753	outputs/results.json · offline.policies.gbm-sensors.c_index	-	✓
F-16	29%	outputs/results.json closed_loop · cox-sensors, crew 6, failures_caught_share (0.2929)	x100, rounded	✓
F-17	69%	outputs/results.json closed_loop · gbm-sensors, crew 6, failures_caught_share (0.6868)	x100, rounded	✓

Assumptions: A-07, A-08

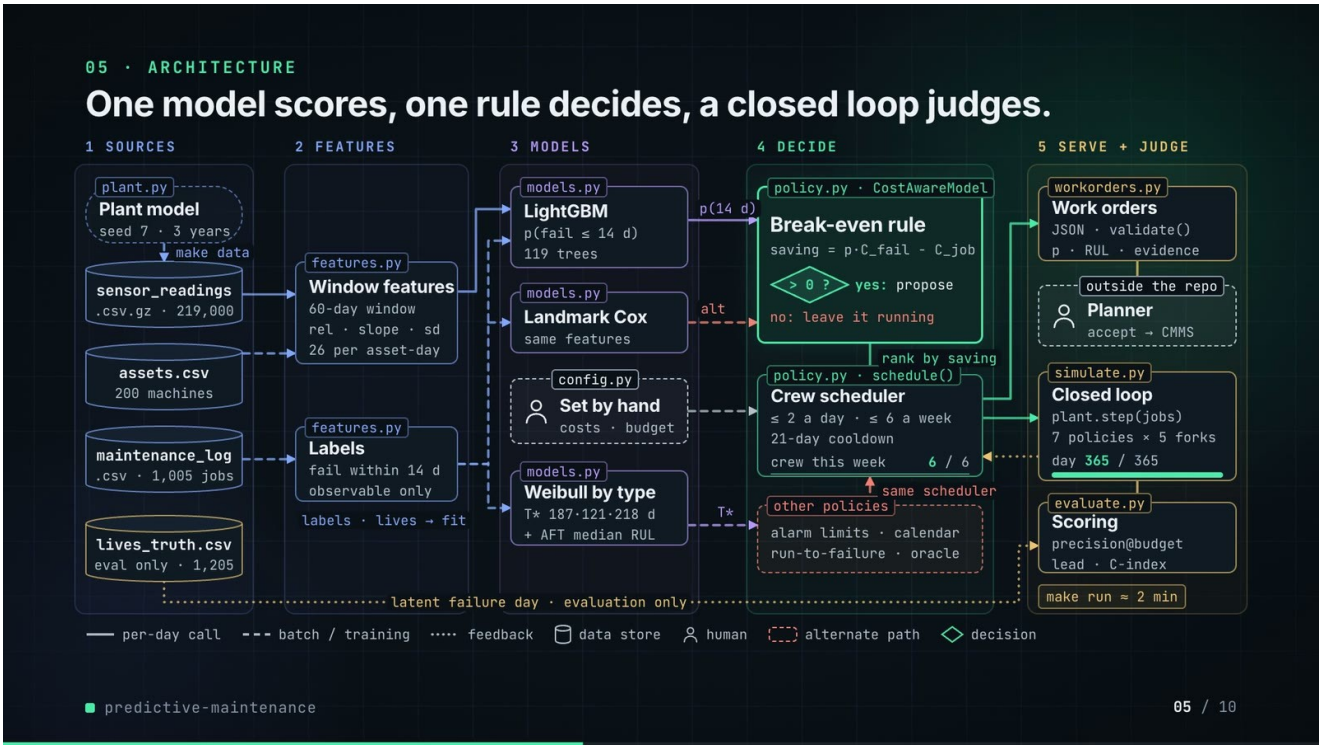
Illustrative, not data:

- The fork drawing is schematic; tick positions are the real failure days of fork 1 (seed 101), scaled to the branch length.

To change this slide: Markup under #ex-b in compositions/explainer.html; ticks come from tools/inject_sheet.py (LOOP block); footnote figures are F-14..F-17.

Slide 05: Architecture

26.0-40.8 s · composition `compositions/s03-architecture.html`



On-screen text (verbatim)

05 · ARCHITECTURE

One model scores, one rule decides, a closed loop judges.

1 SOURCES: plant.py Plant model (seed 7 · 3 years) → make data → sensor_readings.csv.gz (219,000) · assets.csv (200 machines) · maintenance_log.csv (1,005 jobs) · lives_truth.csv (eval only · 1,205)

2 FEATURES: features.py Window features (60-day window · rel · slope · sd · 26 per asset-day) · features.py Labels (fail within 14 d · observable only) · labels · lives → fit

3 MODELS: models.py LightGBM (p(fail ≤ 14 d) · 119 trees) · models.py Landmark Cox (same features) · config.py Set by hand (costs · budget) · models.py Weibull by type (T* 187·121·218 d · + AFT median RUL)

4 DECIDE: policy.py · CostAwareModel Break-even rule (saving = p·C_fail − C_job; > 0 ? yes: propose; no: leave it running) → rank by saving → policy.py · schedule() Crew scheduler (≤ 2 a day · ≤ 6 a week · 21-day cooldown · crew this week 0..6 / 6) · other policies (alarm limits · calendar · run-to-failure · oracle) → same scheduler

5 SERVE + JUDGE: workorders.py Work orders (JSON · validate() · p · RUL · evidence) → Planner (outside the repo · accept → CMMS) · simulate.py Closed loop (plant.step(jobs) · 7 policies × 5 forks · day 0..365 / 365) → evaluate.py Scoring (precision@budget · lead · C-index) · make run ≈ 2 min

latent failure day · evaluation only (lives_truth.csv → Scoring)

critical path: readings → features → p → break-even → crew → work order

Legend: per-day call · batch / training · feedback · data store · human · alternate path · decision

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	200 machines	`data/assets.csv` · row count	-	✓
F-06	219,000 daily sensor readings / .csv.gz · 219,000	`data/sensor_readings.csv.gz` · row count	-	✓
F-18	seed 7 · 3 years	`src/predictive_maintenance/cli.py (--data-seed default 7); config.py:14` · HISTORY_YEARS = 3	-	✓
F-19	.csv · 1,005 jobs	`data/maintenance_log.csv` · row count	-	✓
F-20	eval only · 1,205	`data/lives_truth.csv` · row count	-	✓
F-21	observable only	`src/predictive_maintenance/features.py (_labels); tests/test_features.py:33` · test_labels_drop_rows_a_preventive_job_hides	-	✓
F-22	plant.py, features.py, models.py (LightGBM, Cox, Weibull, AFT), config.py, policy.py (CostAwareModel,	`src/predictive_maintenance/` · classes and functions	-	✓

ID	Shown	Source · key	Calculation	Check
	schedule, other policies), workorders.py, simulate.py, evaluate.py			
F-23	60-day window	src/predictive_maintenance/config.py:19 · WINDOW_DAYS = 60	-	✓
F-24	26 per asset-day	src/predictive_maintenance/features.py:20-28 (FEATURE_NAMES) · 5 sensors x 4 + load_m7 + load_rel3 + days_since_maint + 3 type flags	5 x 4 + 3 + 3 = 26	✓
F-25	p(fail ≤ 14 d) / fail within 14 d / p(14 d)	outputs/results.json · horizon_days	-	✓
F-26	119 trees	outputs/results.json · training.best_iteration	-	✓
F-27	T* 187·121·218 d	outputs/results.json · fixed_intervals_days	-	✓
F-28	saving = p·C_fail − C_job; > 0 ? yes: propose	src/predictive_maintenance/policy.py:109-110 · saving = p * FAILURE_COST - PLANNED_COST; propose where saving > 0	-	✓
F-29	≤ 2 a day · ≤ 6 a week · 21-day cooldown	src/predictive_maintenance/policy.py:144-145; config.py:20, :106-107 · per_day = ceil(6 / 5); COOLDOWN_DAYS = 21	ceil(6/5) = 2	✓
F-30	7 policies × 5 forks	outputs/closed_loop_runs.csv · distinct policies at crew 6; distinct seeds	-	✓
F-31	make run ≈ 2 min / ~2 min	outputs/results.json · runtime_seconds = 127.3	127.3 / 60 = 2.1 min	✓
F-32	Planner · accept → CMMS	src/predictive_maintenance/workorders.py:1-5 (docstring); README 'Known limitations' (schema mapping not done) · -	-	✓

Assumptions: A-01, A-03, A-04, A-05, A-07, A-08

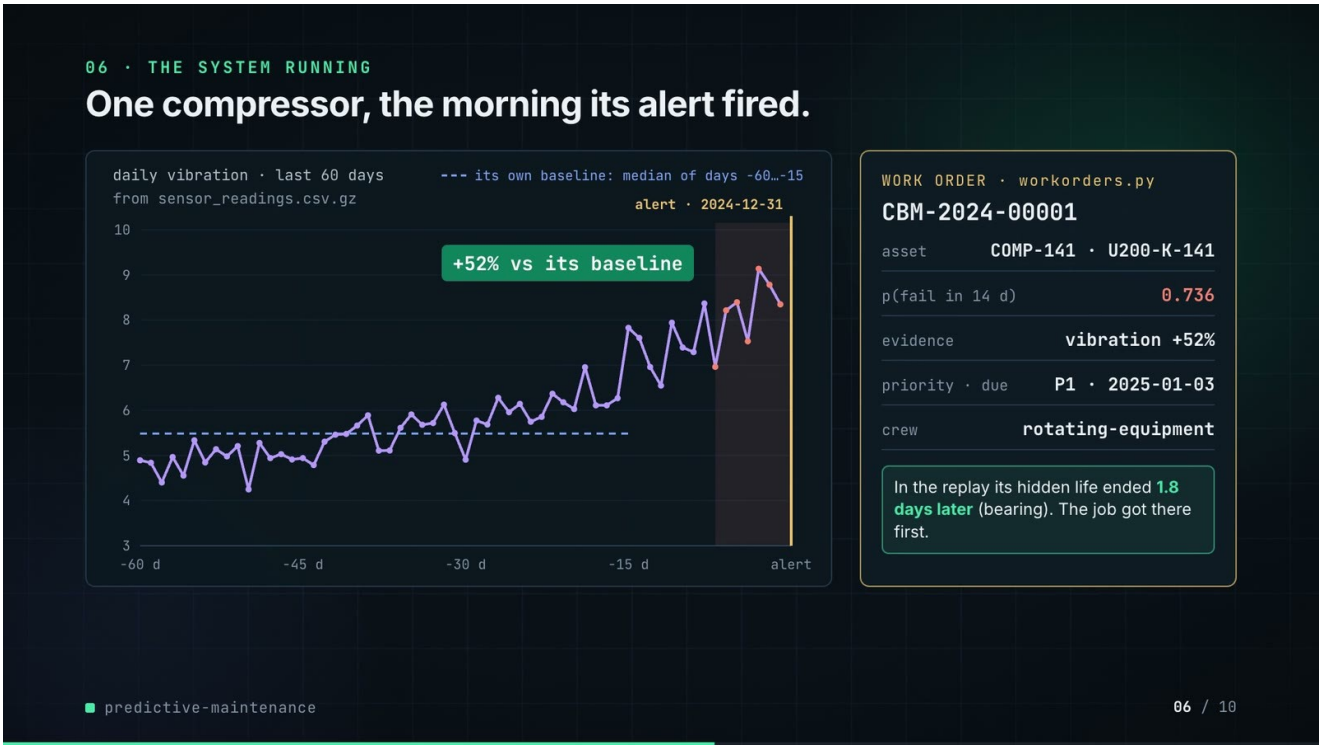
Illustrative, not data:

- Packets moving along the wires are decoration; the day counter and crew bar animate the loop, they are not timings.
- Arrows are the main data flows, not every call (e.g. days since service from the maintenance log into the features is not drawn).

To change this slide: Node boxes (#n-*) and the wires array in compositions/s03-architecture.html.

Slide 06: The system running

40.4-55.0 s · composition `compositions/s04-running.html (RUN block injected by tools/inject_sheet.py)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

One compressor, the morning its alert fired. — daily vibration · last 60 days · from sensor_readings.csv.gz · its own baseline: median of days −60…−15 · +52% vs its baseline · alert · 2024-12-31

WORK ORDER · workorders.py CBM-2024-00001 · asset COMP-141 · U200-K-141 · p(fail in 14 d) 0.736 · evidence vibration +52% · priority · due P1 · 2025-01-03 · crew rotating-equipment

In the replay its hidden life ended 1.8 days later (bearing). The job got there first.

Same plant, same year, six policies. — closed loop · fork 1 of 5 · crew 6 jobs / week · cumulative breakdowns · day 0..365 / 365

run-to-failure 232 · Cox 138 · calendar 111 · alarm limits 56 · model 43 · oracle 3

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-25	p(fail ≤ 14 d) / fail within 14 d / p(14 d)	outputs/results.json · horizon_days	-	✓
F-33	daily vibration · last 60 days (line chart)	data/sensor_readings.csv.gz; demo assets/plant_run.json trace · asset_id = COMP-141, 2024-11-01..2024-12-30	values compared to 3 decimals	✓
F-34	alert · 2024-12-31	outputs/work_orders.json · [0].created_on	-	✓
F-35	+52% vs its baseline / vibration +52%	outputs/work_orders.json · [0].evidence[0] = vibration, 0.521	x100, rounded	✓
F-36	CBM-2024-00001 · COMP-141 · U200-K-141 · 0.736 · P1 · 2025-01-03 · rotating-equipment	outputs/work_orders.json · [0]	-	✓
F-37	hidden life ended 1.8 days later (bearing)	demo assets/plant_run.json work_orders[0].truth (Plant.latent_remaining at the alert, fork seed 101); corroborated by data/lives_truth.csv (the same life, in the history run, ended by a bearing failure on day 731) · latent_days_to_failure, latent_mode	-	✓
F-38	232 · 138 · 111 · 56 · 43 · 3	outputs/closed_loop_runs.csv (seed 101, crew 6); curves from demo assets/plant_run.json · failures	-	✓
F-39	fork 1 of 5	src/predictive_maintenance/config.py:108; demo assets/plant_run.json seed · eval_seeds[0] = 101	-	✓

Assumptions: A-02, A-05, A-07, A-08

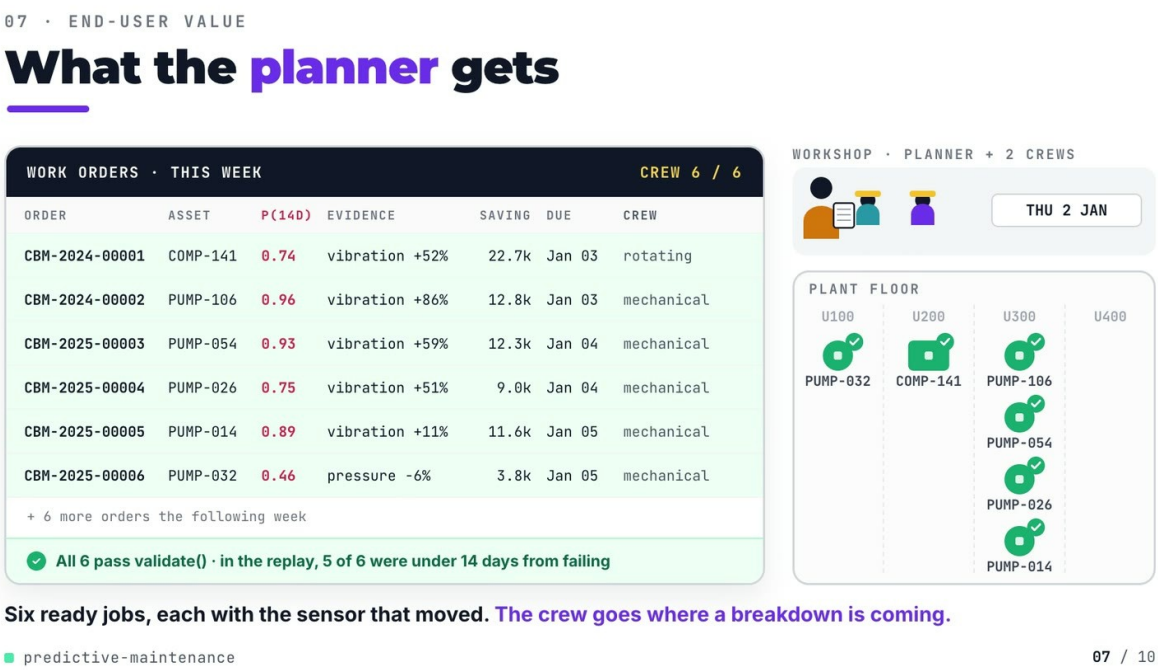
Illustrative, not data:

- Charts are drawn from exported numbers (the repo saves no plots): the trace is read from sensor_readings.csv.gz and the curves are recorded by tools/export_work_orders.py with the repo's own schedule() and Plant.step(); their year-end totals are checked against closed_loop_runs.csv (F-38).
- Baseline line is recomputed in the composition as the window median of days −60…−15, the definition in features.py.

To change this slide: Phase A/B markup and the RUN block in compositions/s04-running.html; re-run export_work_orders.py + inject_sheet.py after make run.

Slide 07: End-user value (white)

54.7-67.7 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE
What the planner gets
WORK ORDERS · THIS WEEK · CREW 6 / 6 — columns ORDER, ASSET, P(14D), EVIDENCE, SAVING, DUE, CREW
CBM-2024-00001 COMP-141 0.74 vibration +52% 22.7k Jan 03 rotating
CBM-2024-00002 PUMP-106 0.96 vibration +86% 12.8k Jan 03 mechanical
CBM-2025-00003 PUMP-054 0.93 vibration +59% 12.3k Jan 04 mechanical
CBM-2025-00004 PUMP-026 0.75 vibration +51% 9.0k Jan 04 mechanical
CBM-2025-00005 PUMP-014 0.89 vibration +11% 11.6k Jan 05 mechanical
CBM-2025-00006 PUMP-032 0.46 pressure -6% 3.8k Jan 05 mechanical
+ 6 more orders the following week · ✓ All 6 pass validate() · in the replay, 5 of 6 were under 14 days from failing
WORKSHOP · PLANNER + 2 CREWS · TUE 31 DEC / WED 1 JAN / THU 2 JAN · PLANT FLOOR U100 U200 U300 U400
Six ready jobs, each with the sensor that moved. The crew goes where a breakdown is coming.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-35	+52% vs its baseline / vibration +52%	`outputs/work_orders.json` · [0].evidence[0] = vibration, 0.521	x100, rounded	✓
F-40	CBM-2024-00001 … CBM-2025-00006	`outputs/work_orders.json [0:6], injected into compositions/value.html` · work_order_id, asset_id, failure_probability, evidence[0], expected_saving, due_by, crew	p to 2 dp; saving / 1000 to 1 dp; crew shortened (rotating-equipment → rotating)	✓
F-41	+ 6 more orders the following week	`outputs/results.json` · work_orders_first_two_weeks = 12	12 - 6	✓
F-42	All 6 pass validate()	`src/predictive_maintenance/cli.py (assert not bad); tools/export_work_orders.py asserts the same` · validate()	-	✓
F-43	5 of 6 were under 14 days from failing	`demo assets/plant_run.json work_orders[0:6].truth` · latent_days_to_failure: 1.8, 0.3, 5.0, 11.0, 3.3, 18.1	count(0 < d <= 14)	✓
F-44	CREW 6 / 6	`outputs/work_orders.json` · orders created 2024-12-30..2025-01-03	count	✓
F-45	TUE 31 DEC · WED 1 JAN · THU 2 JAN	`outputs/work_orders.json` · created_on of orders 1-6	-	✓

Assumptions: A-03, A-04, A-08, A-09

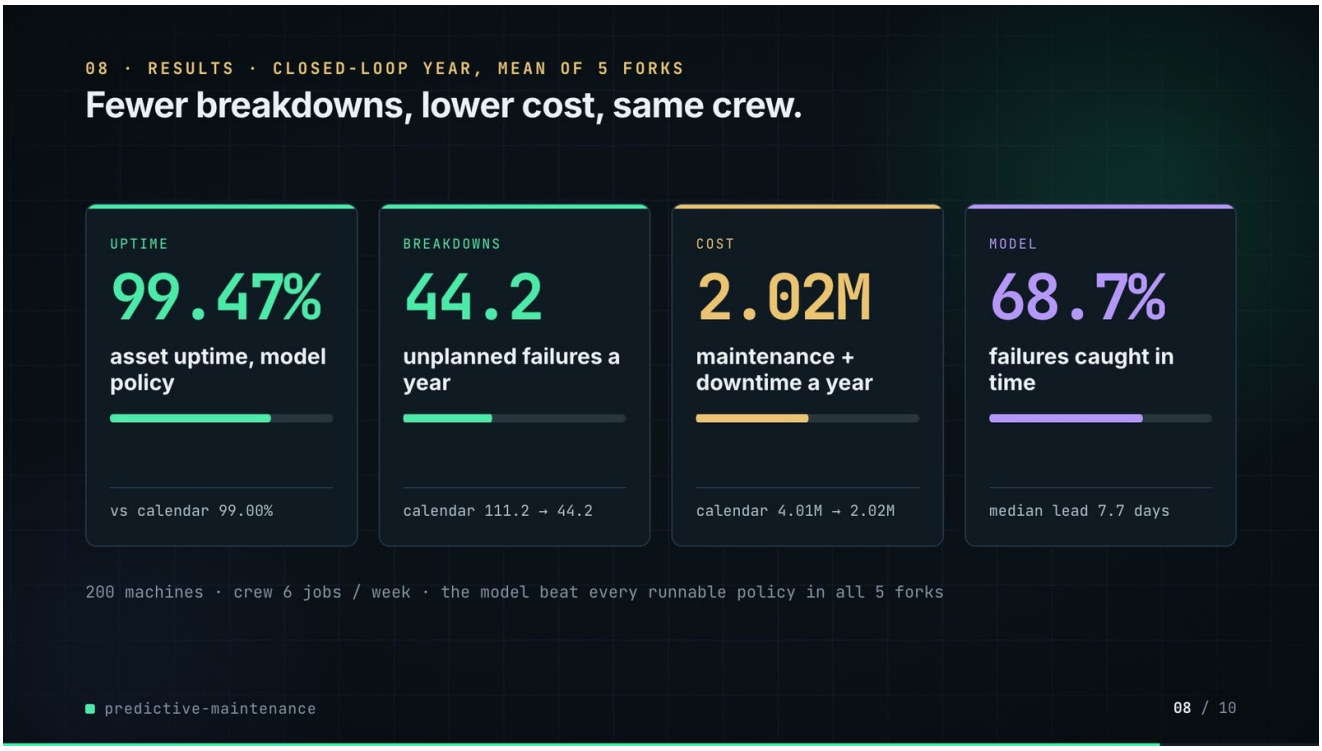
Illustrative, not data:

- Planner, the two crews (the repo has one crew pool with typed crew labels), hard hats, machine drawings and walking speed are illustrative.
- Machines are placed in the unit from their functional location (U100-U400); positions inside a unit are not real.
- SAVING is the model's expected saving (p x failure cost - planned cost), not a realised saving.
- '5 of 6' uses the plant's latent truth, which a real planner would not see.

To change this slide: Re-run tools/export_work_orders.py, then tools/inject_sheet.py (--rows picks the orders); layout in compositions/value.html.

Slide 08: Results

67.4-76.2 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · CLOSED-LOOP YEAR, MEAN OF 5 FORKS

Fewer breakdowns, lower cost, same crew.

UPTIME 99.47% asset uptime, model policy · vs calendar 99.00%

BREAKDOWNS 44.2 unplanned failures a year · calendar 111.2 → 44.2

COST 2.02M maintenance + downtime a year · calendar 4.01M → 2.02M

MODEL 68.7% failures caught in time · median lead 7.7 days

200 machines · crew 6 jobs / week · the model beat every runnable policy in all 5 forks

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	200 machines	`data/assets.csv` · row count	-	✓
F-04	About 6 planned jobs a week / crew 6 jobs / week	`outputs/results.json` · crew_jobs_per_week	-	✓
F-46	99.47%	`outputs/results.json closed_loop` · gbm-sensors, crew 6, uptime	x100	✓
F-47	vs calendar 99.00%	`outputs/results.json closed_loop` · fixed-interval, crew 6, uptime	x100	✓
F-49	44.2	`outputs/results.json closed_loop` · gbm-sensors, crew 6, failures	-	✓
F-50	calendar 111.2	`outputs/results.json closed_loop` · fixed-interval, crew 6, failures	-	✓
F-51	2.02M	`outputs/results.json closed_loop` · gbm-sensors, crew 6, cost_total = 2,015,700	/ 1e6	✓
F-52	4.01M	`outputs/results.json closed_loop` · fixed-interval, crew 6, cost_total = 4,009,400	/ 1e6	✓
F-53	68.7%	`outputs/results.json closed_loop` · gbm-sensors, crew 6, failures_caught_share	x100	✓
F-54	median lead 7.7 days	`outputs/results.json closed_loop` · gbm-sensors, crew 6, median_lead_days	-	✓
F-55	the model beat every runnable policy in all 5 forks	`outputs/results.json` · paired_gbm_minus.*.seeds_gbm_better_uptime	all == 5	✓

Assumptions: A-03, A-04, A-06, A-07, A-09

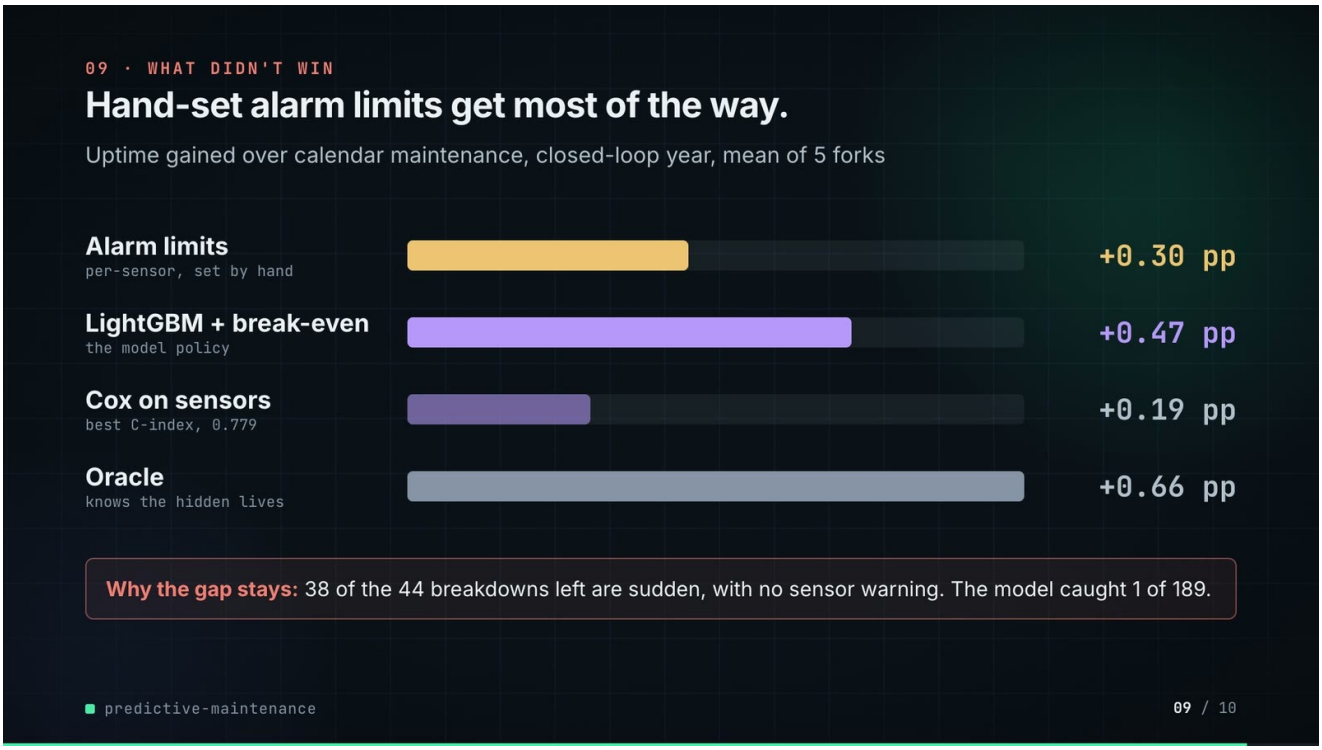
Illustrative, not data:

- Mini bars: uptime bar = model gain / oracle gain over calendar (0.47 / 0.66); failures and cost bars = after / before; caught bar = 68.7%.

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

75.8-82.0 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

Hand-set alarm limits get most of the way.

Uptime gained over calendar maintenance, closed-loop year, mean of 5 forks

Alarm limits (per-sensor, set by hand) +0.30 pp · LightGBM + break-even (the model policy) +0.47 pp · Cox on sensors (best C-index, 0.779) +0.19 pp · Oracle (knows the hidden lives) +0.66 pp

Why the gap stays: 38 of the 44 breakdowns left are sudden, with no sensor warning. The model caught 1 of 189.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-14	0.779	`outputs/results.json` · offline.policies.cox-sensors.c_index	-	✓
F-48	+0.47 pp	`outputs/results.json` · paired_gbm_minus.fixed-interval.uptime_pp_mean	-	✓
F-56	+0.30 pp	`outputs/results.json closed_loop` · alarm-limits uptime - fixed-interval uptime (crew 6)	(0.992978 - 0.989989) x 100 = 0.299	✓
F-57	+0.19 pp	`outputs/results.json closed_loop` · cox-sensors uptime - fixed-interval uptime (crew 6)	(0.991939 - 0.989989) x 100 = 0.195	✓
F-58	+0.66 pp	`outputs/results.json closed_loop` · oracle uptime - fixed-interval uptime (crew 6)	(0.996549 - 0.989989) x 100 = 0.656	✓
F-59	38 of the 44 breakdowns left are sudden	`outputs/closed_loop_runs.csv` · gbm-sensors, crew 6: mean sudden_failures 37.6, mean failures 44.2	rounded to whole numbers	✓
F-60	The model caught 1 of 189	`outputs/results.json` · gbm_caught_by_mode.sudden	-	✓

Assumptions: A-02, A-07, A-09

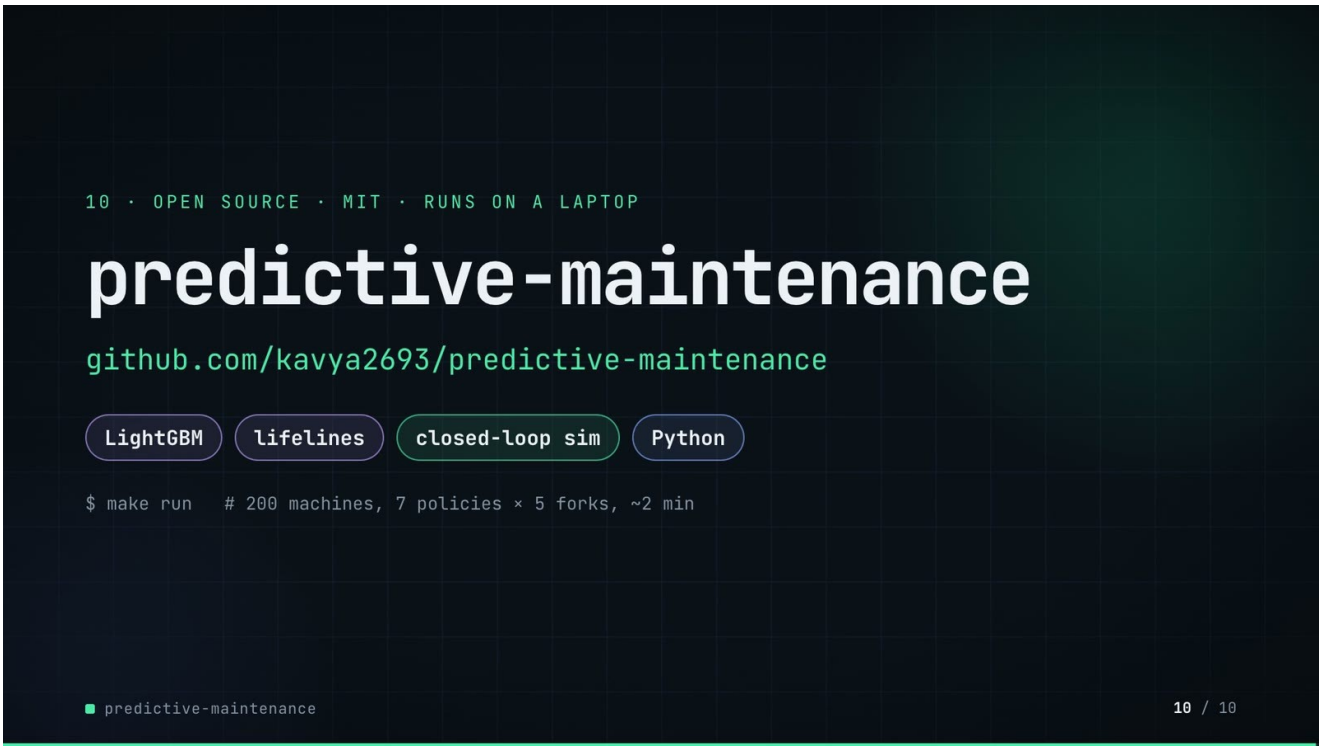
Illustrative, not data:

- Bar lengths: 900 px track = 0.66 pp (the oracle).

To change this slide: Rows #twist-r1..r4 (bar widths are inline styles) in compositions/s06-twist.html.

Slide 10: End card

81.6-86.0 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP

predictive-maintenance

github.com/kavya2693/predictive-maintenance

LightGBM · lifelines · closed-loop sim · Python

$ make run # 200 machines, 7 policies × 5 forks, ~2 min

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-30	7 policies × 5 forks	`outputs/closed_loop_runs.csv` · distinct policies at crew 6; distinct seeds	-	✓
F-31	make run ≈ 2 min / ~2 min	`outputs/results.json` · runtime_seconds = 127.3	127.3 / 60 = 2.1 min	✓
F-61	MIT	`LICENSE line 1` · -	-	✓
F-62	LightGBM · lifelines · closed-loop sim · Python	`pyproject.toml dependencies; src/predictive_maintenance/simulate.py` · lightgbm, lifelines	-	✓
F-63	$ make run	`Makefile` · run target	-	✓

Assumptions: A-09

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic: a seeded day-by-day plant model (200 assets, 3 years) generates the sensor history, the maintenance log and the latent lives	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/predictive_maintenance/config.py`; `src/predictive_maintenance/plant.py`	make data; make run; export_work_orders.py + inject_sheet.py
A-02: Failure precursors: each failure mode drifts the sensors for a mode-specific number of days before the failure (bearing 21 d, fouling 35 d, overheating 4 d,	F-17, F-34, F-35, F-37, F-53, F-54, F-59, F-60	04, 06, 07, 08, 09	`src/predictive_maintenance/config.py:93-101`	make data; make run; export_work_orders.py + inject_sheet.py
A-03: Costs: a planned job costs 1,500 plus one day of downtime; a failure costs its repair (9k pump, 22k compressor, 12k heat exchanger) plus 3-5 days of downtime at 3,000 a day	F-28, F-40, F-51, F-52	05, 07, 08	`src/predictive_maintenance/config.py:32-34 and the AssetType blocks`	make run; export_work_orders.py + inject_sheet.py
A-04: Crew budget: 6 planned jobs a week, Monday to Friday, at most ceil(6/5) = 2 a day; an asset serviced in the last 21 days is not offered again	F-04, F-29, F-44, F-45, F-46, F-49, F-51, F-53	02, 05, 07, 08	`src/predictive_maintenance/config.py:106, :20`	make run; export_work_orders.py + inject_sheet.py
A-05: Prediction horizon 14 days, feature window 60 days, baseline = window median excluding the last 14 days	F-23, F-25, F-34, F-35	05, 06, 07	`src/predictive_maintenance/config.py:18-19`; `src/predictive_maintenance/features.py:18`	make run; export_work_orders.py + inject_sheet.py
A-06: Every job is a perfect repair (the asset is as good as new); imperfect repair and infant mortality are not modelled	F-46, F-49, F-51, F-53	08	`src/predictive_maintenance/plant.py`	make run
A-07: Closed-loop evaluation: the plant is replayed to the end of the 2-year training history (2024-12-31), forked with seeds 101-105, and each policy runs 365 days from the same fork with the same random stream	F-11, F-12, F-13, F-30, F-38, F-39, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-56, F-57, F-58, F-59	04, 05, 06, 08, 09, 10	`src/predictive_maintenance/config.py:108`	make run; export_work_orders.py + inject_sheet.py
A-08: Truth for scoring: an alert is a catch when the asset's latent life would have ended within 14 days; the latent failure day comes from the plant model (lives_truth	F-16, F-17, F-37, F-43, F-53, F-60	04, 06, 07, 08, 09	`src/predictive_maintenance/simulate.py`	make run
A-09: Video rounding: uptime to 2 decimals, percentages to whole numbers or 1 decimal as shown, costs in millions to 2 decimals, savings in thousands to 1 decimal, failure probability to 2 decimals on the sheet (3 on the work-order card), failure means to 1 decimal or whole numbers as shown	F-07, F-16, F-17, F-31, F-35, F-40, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-56, F-57, F-58, F-59	02, 04, 05, 06, 07, 08, 09, 10	`the composition file named on each slide`	hyperframes render
A-10: Slide 01 (industries, everyday problems) is general maintenance knowledge used as context; it is not a result of this repo and carries no numbers	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

- The biggest gap (README): the same plant model generated the training data and runs the evaluation, so every result says the method works on a plant with these physics, not what it would achieve on a given real plant.
- Slides 04 and 06 show one fork (seed 101); slides 08-09 show the mean of five forks. Seed-101 totals (e.g. model 43 failures) differ from the 5-fork means (44.2).
- 'Hidden life ended 1.8 days later' and '5 of 6 under 14 days from failing' use the plant's latent truth (Plant.latent_remaining), which exists only in simulation.
- Expected saving on the sheet is the model's own estimate (p x failure cost - planned cost); it is not a realised saving.
- The repo saves no plots; slide 06 charts are drawn from exported numbers (F-33, F-38), not from matplotlib output.
- '+0.47 pp' is the paired mean from results.json; it equals the difference of the two 5-fork mean uptimes.
- Precision at budget (46%) is not on screen: half the model's jobs land on an asset that would not have failed inside 14 days (README 'What disappoints').

- Slide 01 statements are general industry knowledge (A-10).

Appendix: environment

Item	Value
Commit	`70531ff3190a5ff2de81b5edc3ef1ee5888ae47c`
Python	3.11.7
lightgbm	4.7.0
lifelines	0.30.3
pandas	2.3.3
numpy	2.4.6
sklearn	1.9.1
Profile / seeds	data_seed=7, closed_loop_seeds=101-105 (5 forks), fork_shown_on_slides_04_06=101, lightgbm_random_state=0
Run time	results.json runtime_seconds = 127.3 s (about 2 min) for make run on the author's laptop